

MATH 215 SI

Session 4

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Lectures 12 - 17

- ▶ 14.5 The Chain Rule
- ▶ 14.6 Directional Derivatives and the Gradient Vector
- ▶ 14.7 Maximum and Minimum Values
- ▶ 14.8 Lagrange Multipliers

14.5 The Chain Rule

Case 1: Suppose $z = f(x, y)$ is differentiable, and $x = g(t)$, $y = h(t)$ are differentiable functions of t . Then

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

Case 2: Suppose $z = f(x, y)$ is differentiable, and $x = g(s, t)$, $y = h(s, t)$. Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \qquad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

Exercises

14.5 The Chain Rule

- Let $z = x^2y$ where $x = 3t$ and $y = t^2 + 1$. Use the chain rule to find $\frac{dz}{dt}$ when $t = 1$

14.6 Directional Derivatives and the Gradient Vector

The directional derivative of f at (x_0, y_0) in the direction of a **unit** vector $\mathbf{u} = \langle a, b \rangle$ is

$$D_{\mathbf{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

- ▶ The gradient of f is $\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle$
- ▶ If f is differentiable, then $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$
- ▶ If the direction is given by a non-unit vector $\mathbf{v}$, use $\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|}$
- ▶ $D_{\mathbf{i}}f = f_x$ and $D_{\mathbf{j}}f = f_y$

14.6 Directional Derivatives and the Gradient Vector

cont.

- ▶ f increases fastest in the direction of ∇f (at rate $|\nabla f|$) and decreases fastest in the direction of $-\nabla f$
- ▶ $D_{\mathbf{u}}f = 0$ when $\mathbf{u} \perp \nabla f$, i.e. along the level curve

Exercises

14.6 Directional Derivatives and the Gradient Vector

You are walking on the graph $f(x, y) = 2y \cos(\pi x) - 2y \cos(\pi y) + 5$ and you are standing at the point $(1, 1, 4)$

- ▶ Find a direction you should walk in order to stay at a height of 4. Report your answer as a unit vector
- ▶ Find a direction you should walk in order to decrease your height the fastest. Report your answer as a unit vector

14.7 Maximum and Minimum Values

- ▶ f has a local maximum at (a, b) if $f(x, y) \leq f(a, b)$ for all (x, y) near (a, b) (similarly for local minimum)
- ▶ If f has a local extremum at (a, b) and the partials exist there, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$
- ▶ (a, b) is a critical point if $f_x(a, b) = f_y(a, b) = 0$, or if one of the partials does not exist

14.7 Maximum and Minimum Values

cont.

Second Derivatives Test

Suppose (a, b) is a critical point and let

$$D = D(a, b) = f_{xx}(a, b) f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

- ▶ If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum
- ▶ If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum
- ▶ If $D < 0$, then (a, b) is a saddle point
- ▶ If $D = 0$, the test is inconclusive

14.7 Maximum and Minimum Values

cont.

Absolute Maximum and Minimum Values

- ▶ Extreme Value Theorem: if f is continuous on a closed, bounded set D , then f attains an absolute maximum and minimum on D

To find them:

1. Find the values of f at the critical points of f in D
2. Find the extreme values of f on the boundary of D
3. The largest value from steps 1 and 2 is the absolute maximum, the smallest is the absolute minimum

Exercises

14.7 Maximum and Minimum Values

- Find and classify the critical points for the function $h(x, y) = x^4 + y^3 - 6y - 2x^2$

14.8 Lagrange Multipliers

To find the extreme values of $f(x, y, z)$ subject to the constraint $g(x, y, z) = k$ (assuming they exist and $\nabla g \neq \mathbf{0}$ on the constraint):

1. Find all x , y , z , and λ such that

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) \quad g(x, y, z) = k$$

2. Evaluate f at all the points found in step 1. The largest is the maximum, the smallest is the minimum
- Geometrically, at an extremum the level curve/surface of f is tangent to the constraint, so $\nabla f \parallel \nabla g$

14.8 Lagrange Multipliers

cont.

Two Constraints

- To find extreme values of $f(x, y, z)$ subject to $g(x, y, z) = k$ and $h(x, y, z) = c$, solve

$$\nabla f = \lambda \nabla g + \mu \nabla h \quad g(x, y, z) = k \quad h(x, y, z) = c$$

Combining with 14.7

- To find absolute extrema on a closed region, use Lagrange multipliers to handle the boundary, and compare with the critical points in the interior

Exercises

14.8 Lagrange Multipliers

- ▶ Find the maximum and minimum values of the function $f(x, y) = x + y$ on the curve $x^2 + y^2 - xy = 4$
- ▶ Consider a hypothetical planet whose surface can be approximated by the sphere $x^2 + y^2 + z^2 = 1$. Suppose the planet's temperature is strange and given by $T(x, y, z) = 2x + 2y + z$. Find the maximum and minimum temperatures on the planet's surface

Fin

This concludes session 4.